Representing inputs as vectors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Representing inputs as vectors before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

ML Builder Expo

Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A neural network receives numbers in a specified order. Before learning how it computes, we need to represent one example without losing the meaning of its fields.

A **vector** is an ordered list: [4, 2] can mean width 4 cm and height 2 cm if that is the declared order. Reversing the two numbers describes a different rectangle under the same **schema**.

For this fictional parcel investigation, use [width in cm, height in cm]. Width and height must both be present and greater than zero. The **vector** has **dimension** 2 because it contains two **component**s.

Record IDs identify examples; they do not measure the parcel. The target class is stored separately.

Record	Width and height in cm	Input vector
A	4 and 2	[4, 2]
B	2 and 4	[2, 4]
C	3 and 3	[3, 3]

There are three examples, each with two features. Stacking their **vector**s makes a table with three rows and two columns. Do not confuse the number of rows with the **dimension** of one **input**.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Representing inputs as vectors trace and record the first step you can explain without help.

Use W2: Convert before encoding for Representing **input**s as **vector**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Representing inputs as vectors explanation against the supplied facts before discussing it.

Use the worked case for Representing **input**s as **vector**s. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new schema independently

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

W1 Encode the records

> Encode the records. Apply the supplied Representing inputs as vectors facts and keep your reasoning.

Write A, B and C as ordered pairs from the lesson. State the **dimension** of each and the shape of the stacked table.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Convert before encoding

> Convert before encoding. Apply the supplied Representing inputs as vectors facts and keep your reasoning.

A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

> **My reasoning and evidence**

WORKBOOK · 7 OF 15

W3 Handle missing information

> Handle missing information. Apply the supplied Representing inputs as vectors facts and keep your reasoning.

Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this **schema**.

> **My reasoning and evidence**

WORKBOOK · 8 OF 15

W4 Apply a new schema independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new schema independently. Apply the supplied Representing inputs as vectors facts and keep your reasoning.

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Explain a collision

> Explain a collision. Apply the supplied Representing inputs as vectors facts and keep your reasoning.

Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two **input**s apart.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Representing inputs as vectors to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled representing **input**s as **vector**s **evidence** sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Representing inputs as vectors

> Encode the same pixel measurements in a fixed vector order. Explain the labelled step before checking the changed condition.

Encode the same pixel measurements in a fixed **vector** order.

Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/1

Representing inputs as vectors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare coordinate order

Coordinate 0 is left brightness; coordinate 1 is right brightness.

2. Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.

3. Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

CHANGE ONE THING

Swap only A's vector order before prediction.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Encode the same pixel measurements in a fixed **vector** order.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use the same order when **training** and predicting; record missing measurements rather than guessing.

1. Declare coordinate order

2. Check **input** contract

3. Encode both images

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check **input** contract Use the same order when **training** and predicting; record missing measurements rather than guessing.

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Swap only A's **vector** order before **prediction**. Which conclusion is supported by the supplied facts?

1. The encoded **input** becomes [7, 2], numerically identical to B's original **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Repair a missing building block

Review ML Builder Expo (ai-4/36). Goal: Defend the project using traceable **evidence**. Short practice: Which three supporting items should be visible at the expo?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

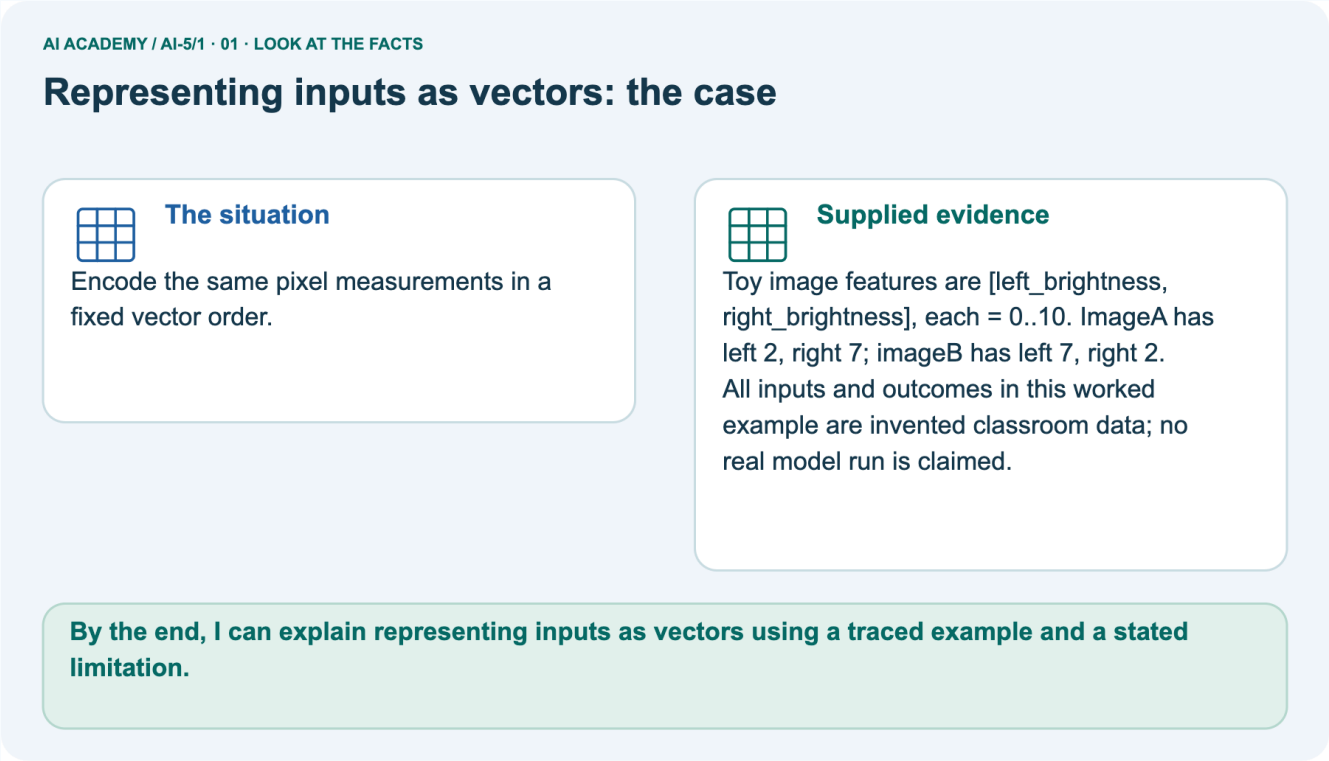
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Representing inputs as vectors picture from its labels, then record a relevant limit.

01. Representing inputs as vectors: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

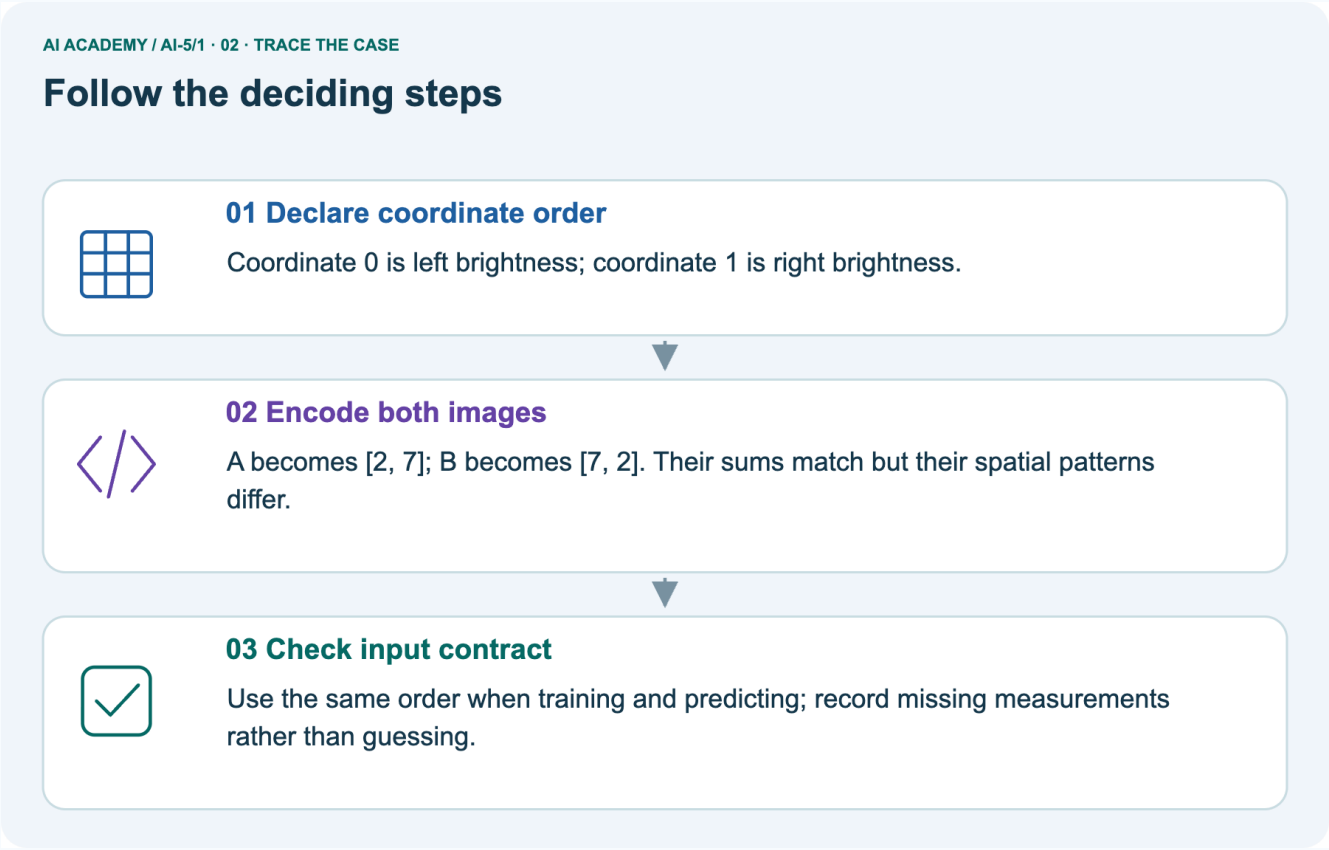
Encode the same pixel measurements in a fixed **vector** order. Supplied facts: Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

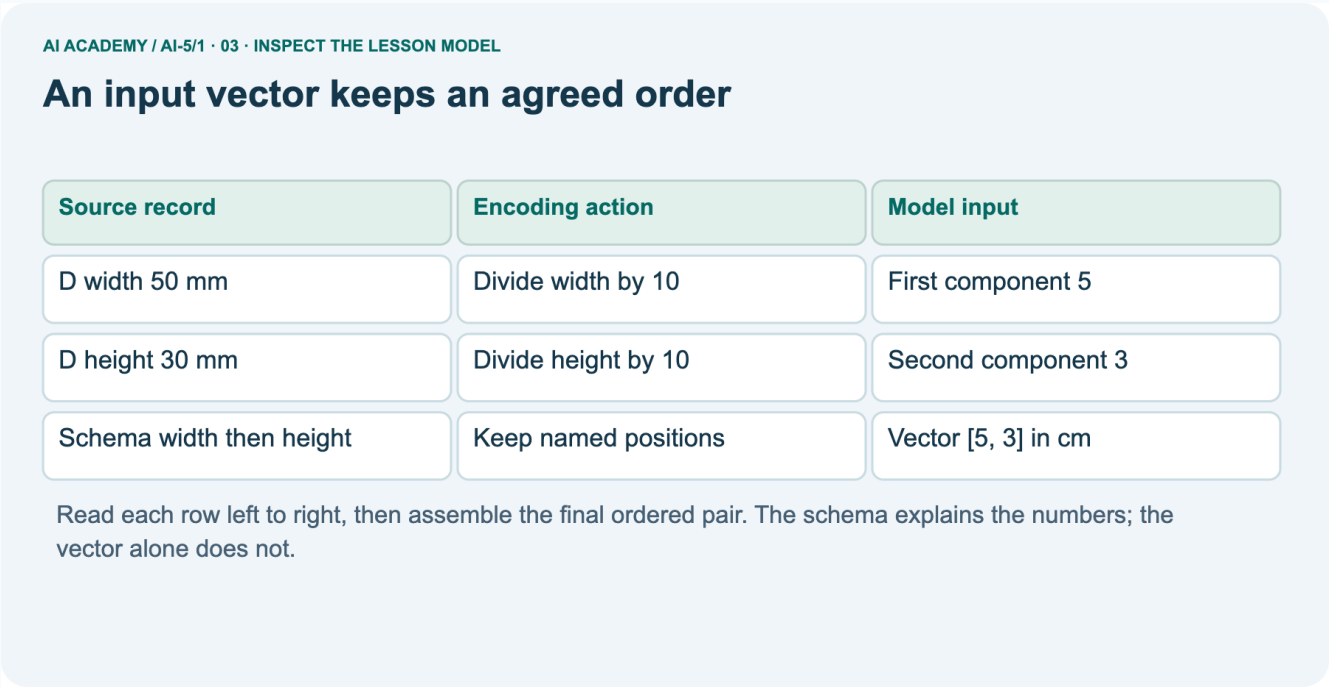
Declare coordinate order: Coordinate 0 is left brightness; coordinate 1 is right brightness. → Encode both images: A becomes [2, 7]; B becomes [7, 2].

Their sums match but their spatial patterns differ. → Check **input** contract: Use the same order when **training** and predicting; record missing measurements rather than guessing.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. An input vector keeps an agreed order



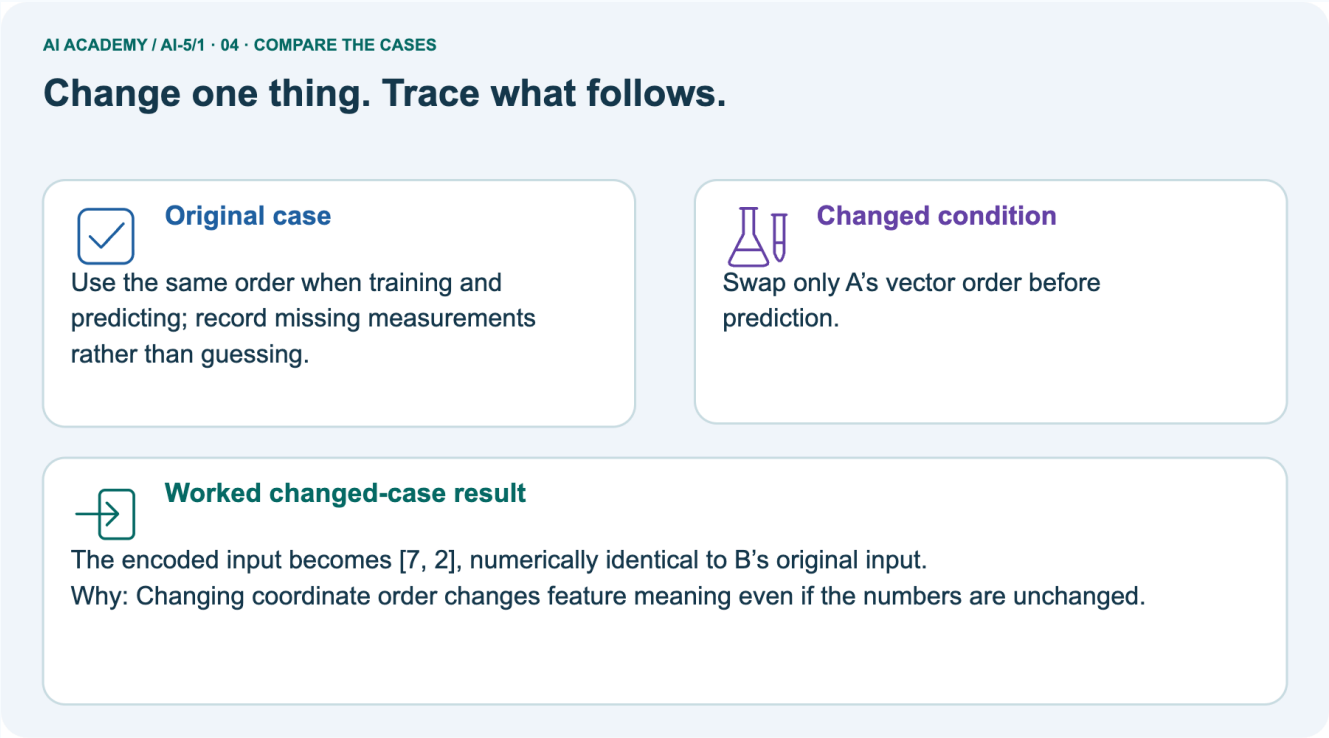
Follow the labelled picture with the matching explanation. Open the vector diagram

An **input vector** keeps an agreed order. D width 50 mm | Divide width by 10 | First **component** 5. D height 30 mm | Divide height by 10 | Second **component** 3. **Schema** width then height | Keep named positions | **Vector** [5, 3] in cm

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

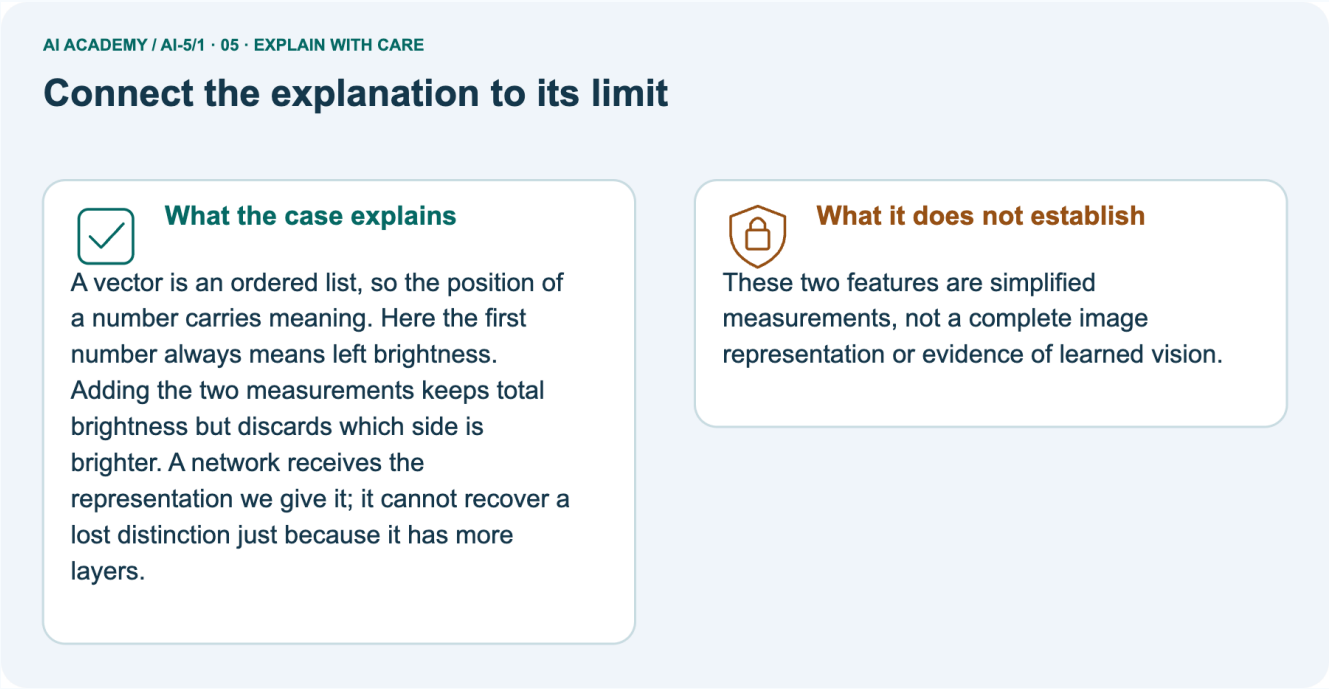
Original case: Use the same order when **training** and predicting; record missing measurements rather than guessing. Changed condition: Swap only A's **vector** order before **prediction**.

Expected result: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

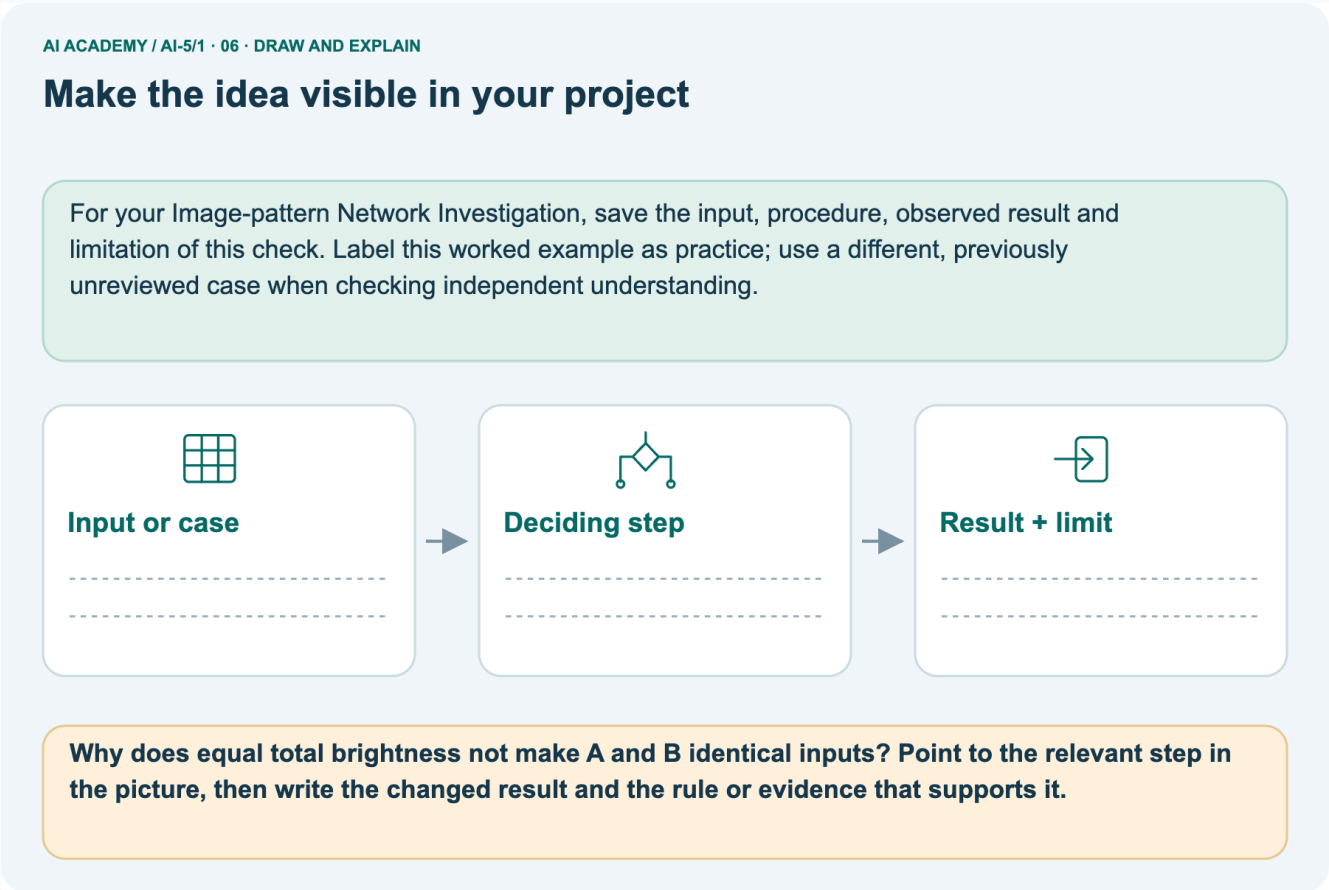
A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Limit: These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.